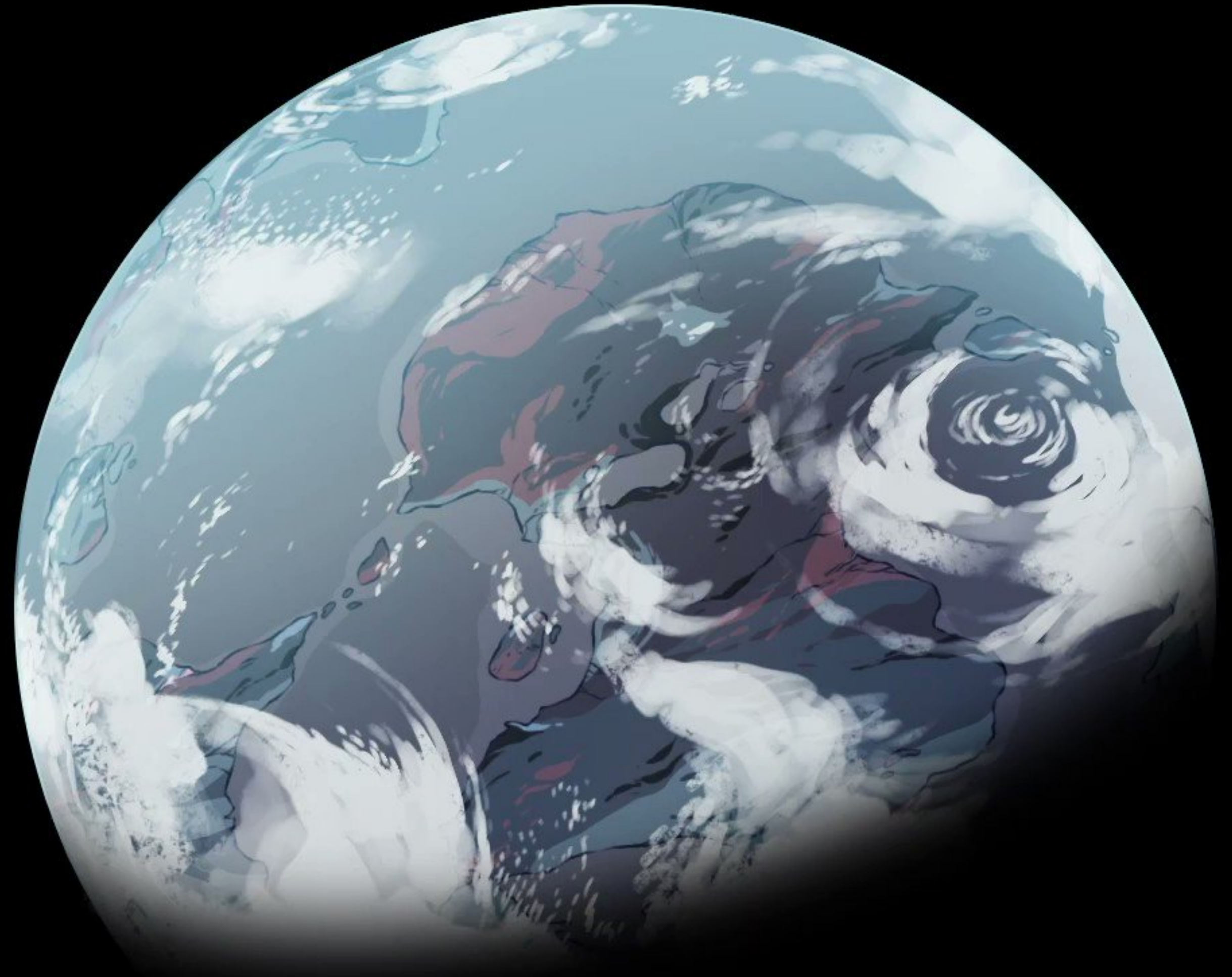


Science & Science Fiction

33-120 | Prof. Katie Breivik



- ▶ Quiz 3: due **tonight** at midnight (85 pts!)
- ▶ Exploration paper 3 is available
 - ▶ due Wednesday, March 19 @ midnight
 - ▶ Can use Chat GPT — see Canvas for details

Midterm grades are in! Please email me if you have questions!

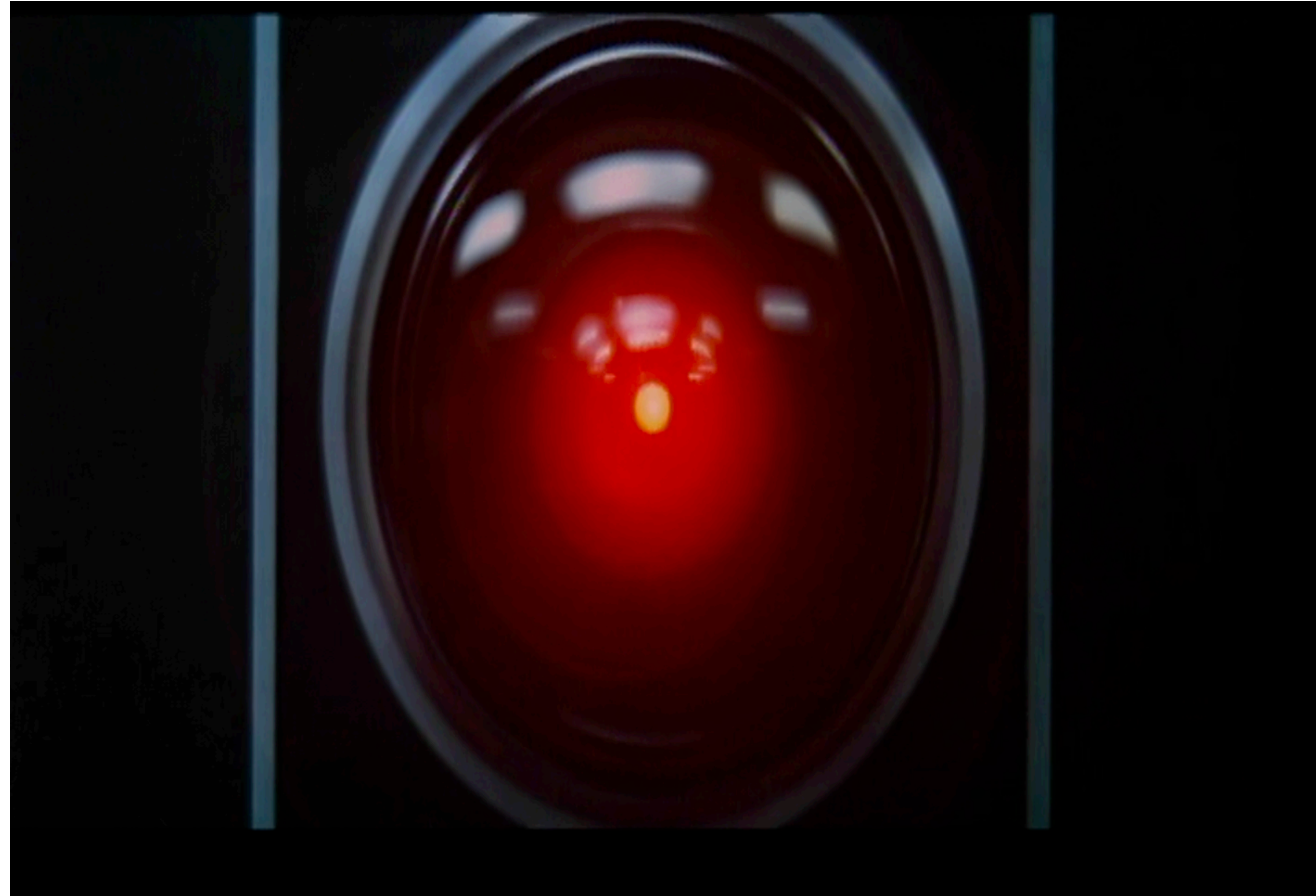
- ▶ Today: The Turing Test

Week 8	03/03/25 No Class: Spring Break	03/05/25 No Class: Spring Break	03/07/25 No Class: Spring Break
Week 9	03/10/25 Intelligent robots	03/12/25 The Turing Test Quiz 3: Matter and Energy (85 pts)	03/14/25 Asimov's Laws and robot behavior
Week 10	03/17/25 Artificial consciousness & robot ethics	03/19/25 Are we alone in the Universe? Conditions necessary for life to exist Exploration Paper 3: AI (25 pts)	03/21/25 Crewed/un-crewed spaceflight Finding habitable planets
Week 11	03/24/25 Finding habitable planets continued Quiz 4: AI (50 pts)	03/26/25 The science behind the SETI project	03/28/25 No class: SETI Team Project Exploration Paper 4: SETI (50 pts)
Week 12	03/31/25 What does it mean to be human? Insight from Frankenstein Medical technology Quiz 5: Extraterrestrial Intelligence (30 pts)	04/02/25 Cell structure and radiation damage DNA, genome sequencing & gene editing SETI Team Project Due (25 pts)	04/04/25 No Class: Spring Carnival
Week 13	04/07/25 Human teleportation	04/09/25 What can we learn from an android about what it means to be human?	04/11/25 How can we solve our problems Public perception of science The scientific method Exploration Paper 5: Being Human (25 pts)
Week 14	04/14/25 Misplaced faith in science Misunderstanding of science Quiz 6: Being Human (50 pts)	04/16/25 Misuse of science in advertising Final Film Critique (50 pts)	04/18/25 What lies ahead? Examples of sci-fi predicting the future How sci-fi storytelling changes with time
Week 15	04/21/25 No class	04/23/25 In class critiques Quiz 7: Problem solving (30 pts)	04/25/25 In class critiques Last day of class – closing discussion

3 requirements for a machine to qualify as an *intelligent robot*

1. A sensor that gathers data autonomously from its environment
2. Computation that makes decisions based on input from the sensor
3. A response that manipulates the environment

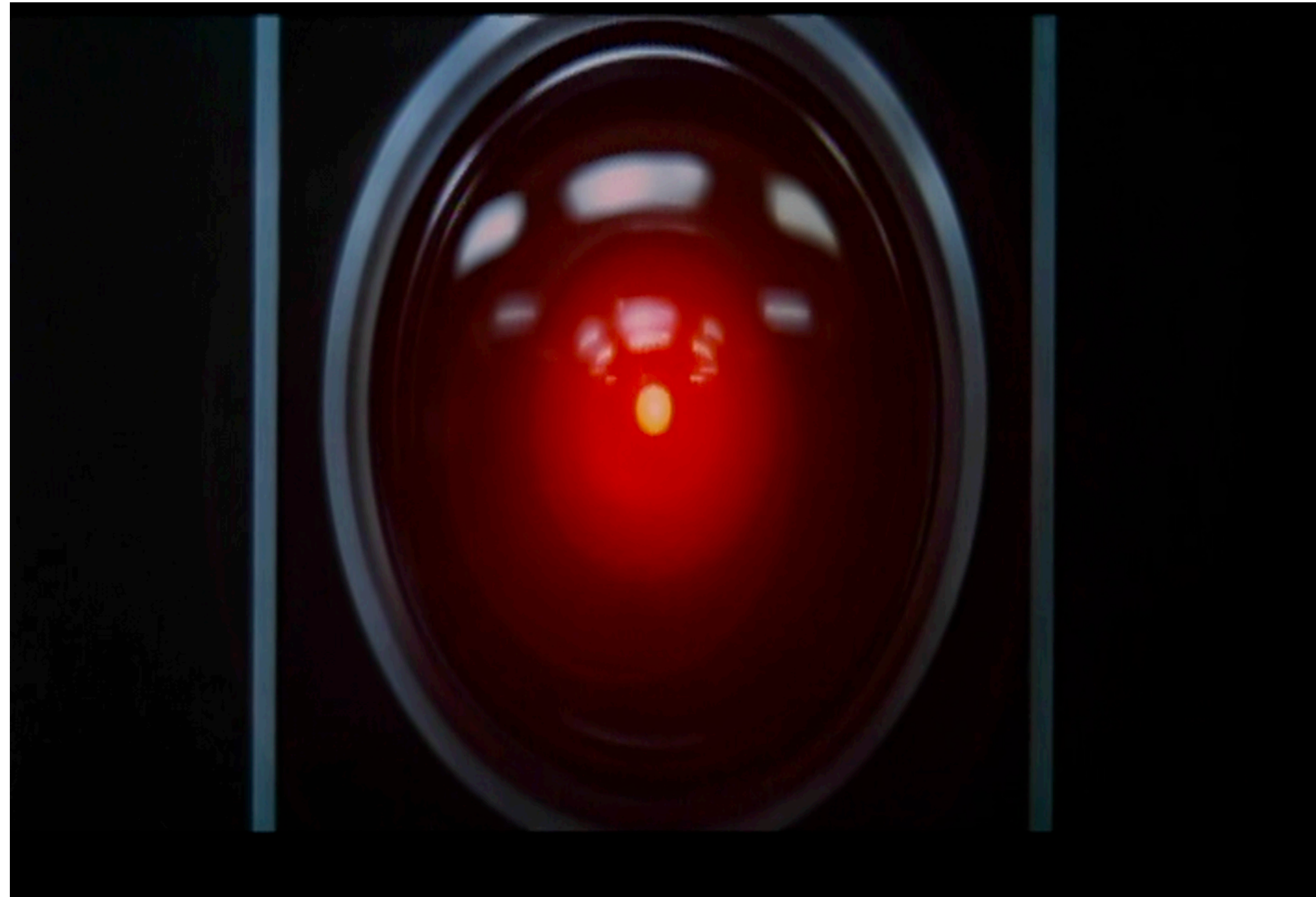




2001: A Space Odyssey

Directed:
Stanley Kubrick

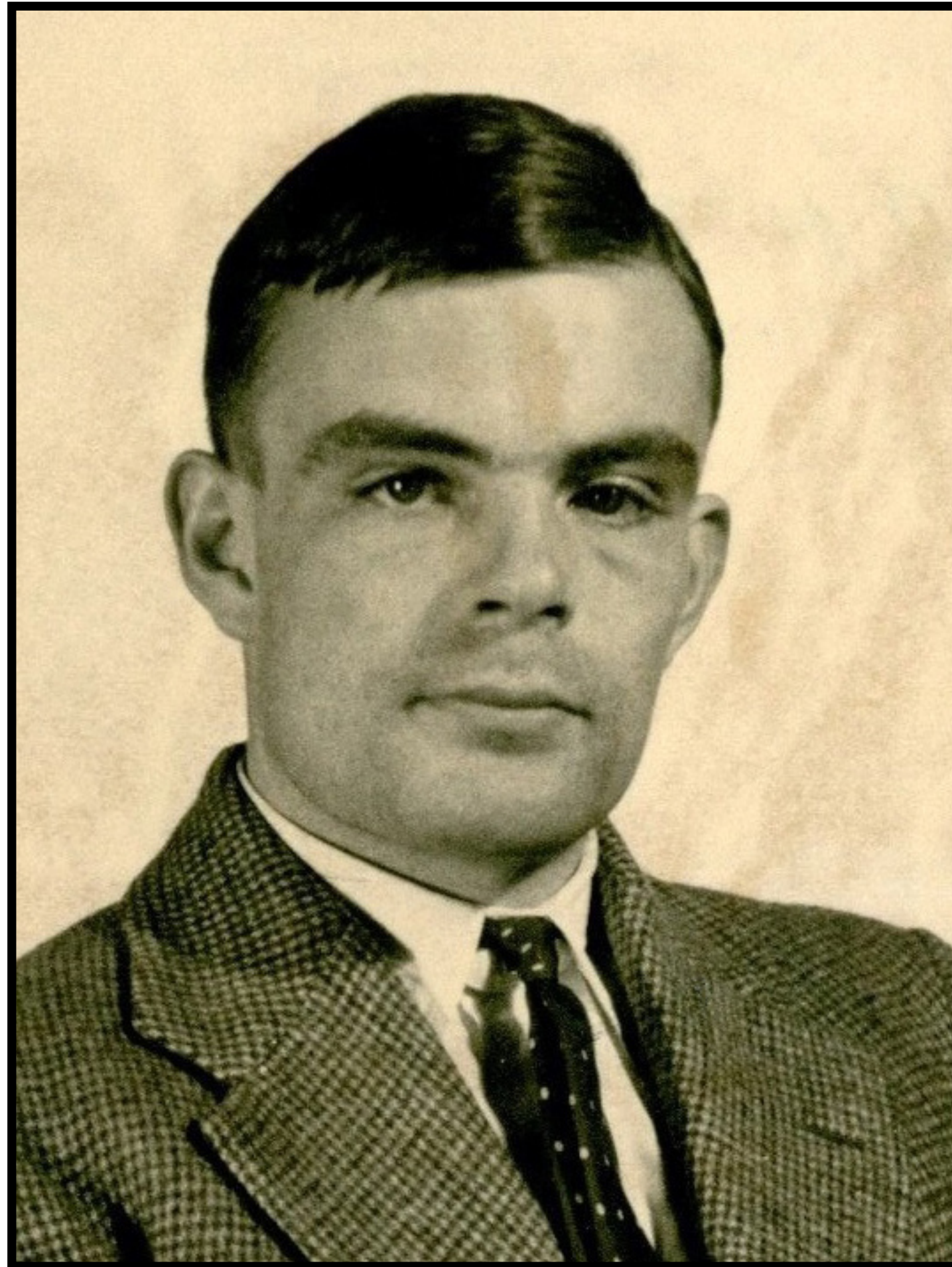
MGM (1968)



3 requirements for a machine to qualify as an *intelligent robot*

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Could HAL convince us that he is actually human?



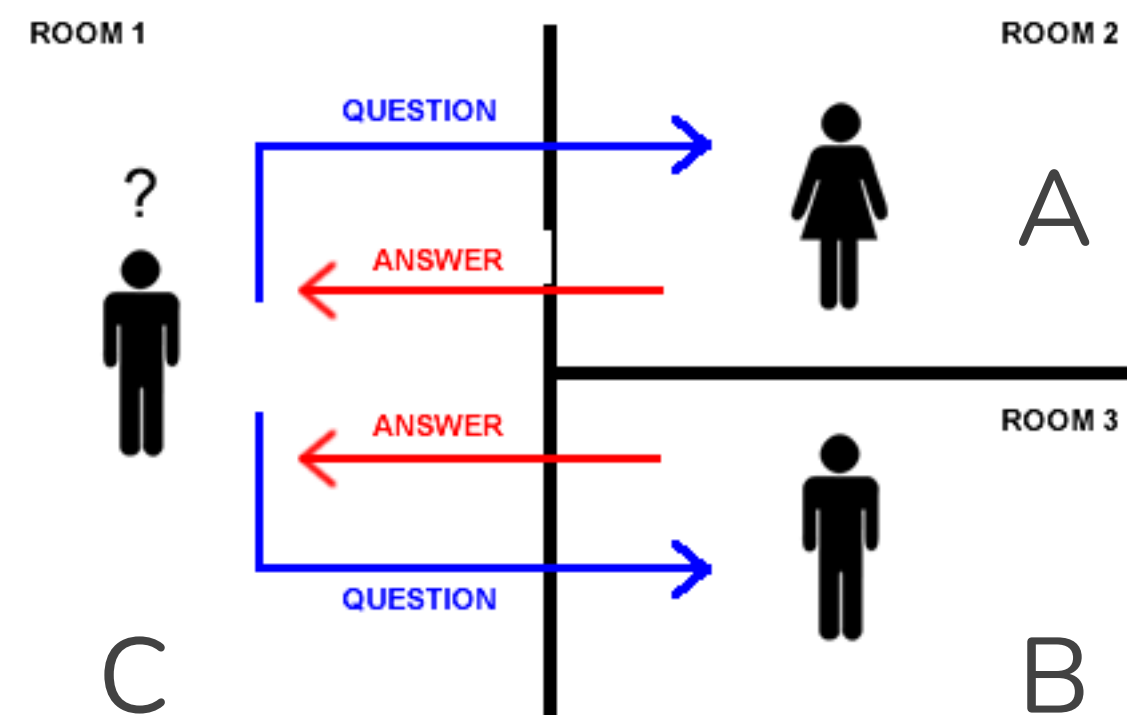
Alan Turing

Computing Machinery and Intelligence

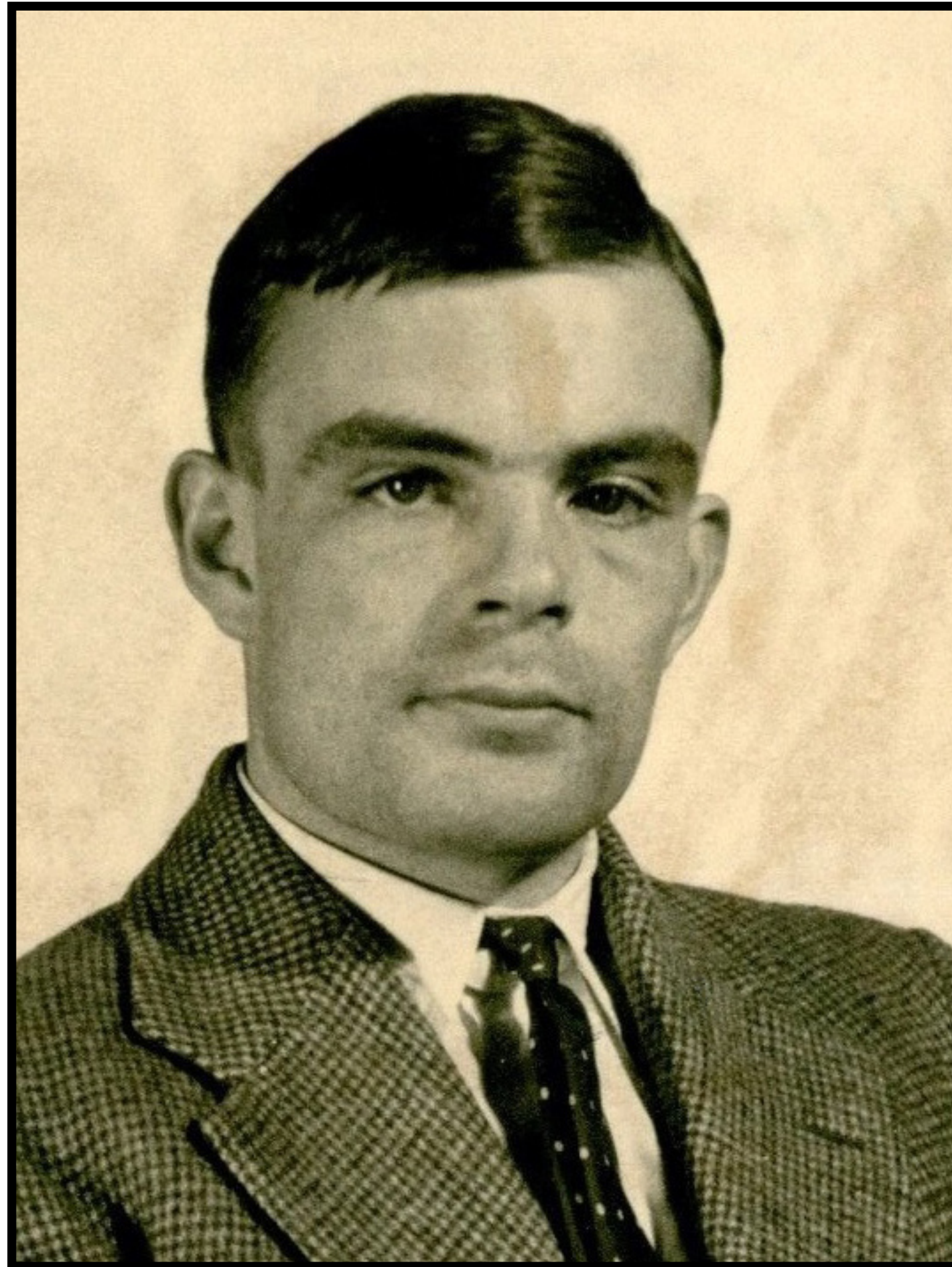
submitted to *Mind* in 1950

I propose to consider the question, "Can machines think?"

The Imitation Game:



Can person C guess whether person A or B is a woman from written text only? How often if this is done **statistically**?



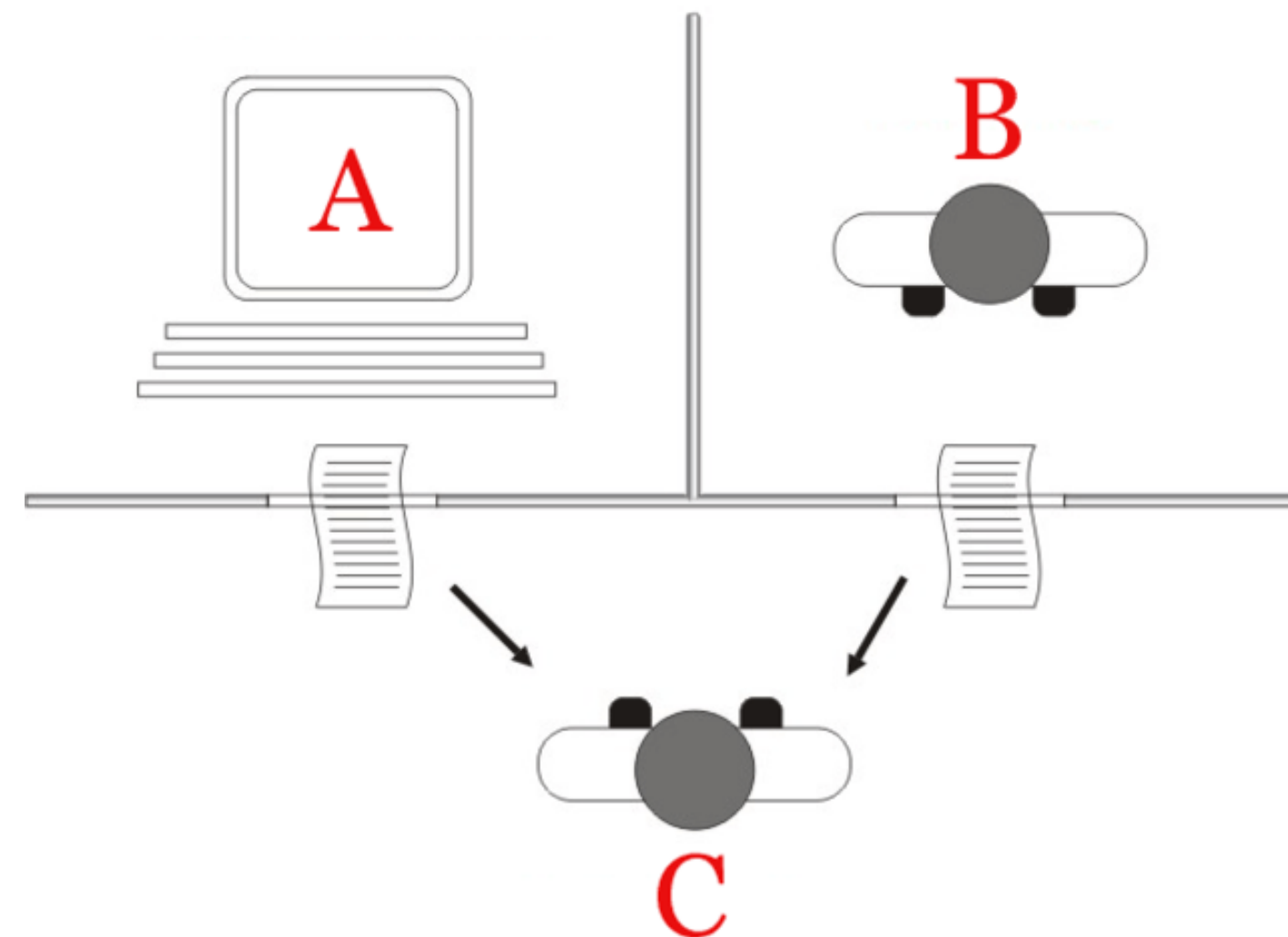
Alan Turing

Computing Machinery and Intelligence

submitted to *Mind* in 1950

I propose to consider the question, "Can machines think?"

The Turing Test:



Can person C guess whether A or B is a *computer* from written text only? How often if this is done **statistically**?



Ex Machina

Directed:
Alex Garland

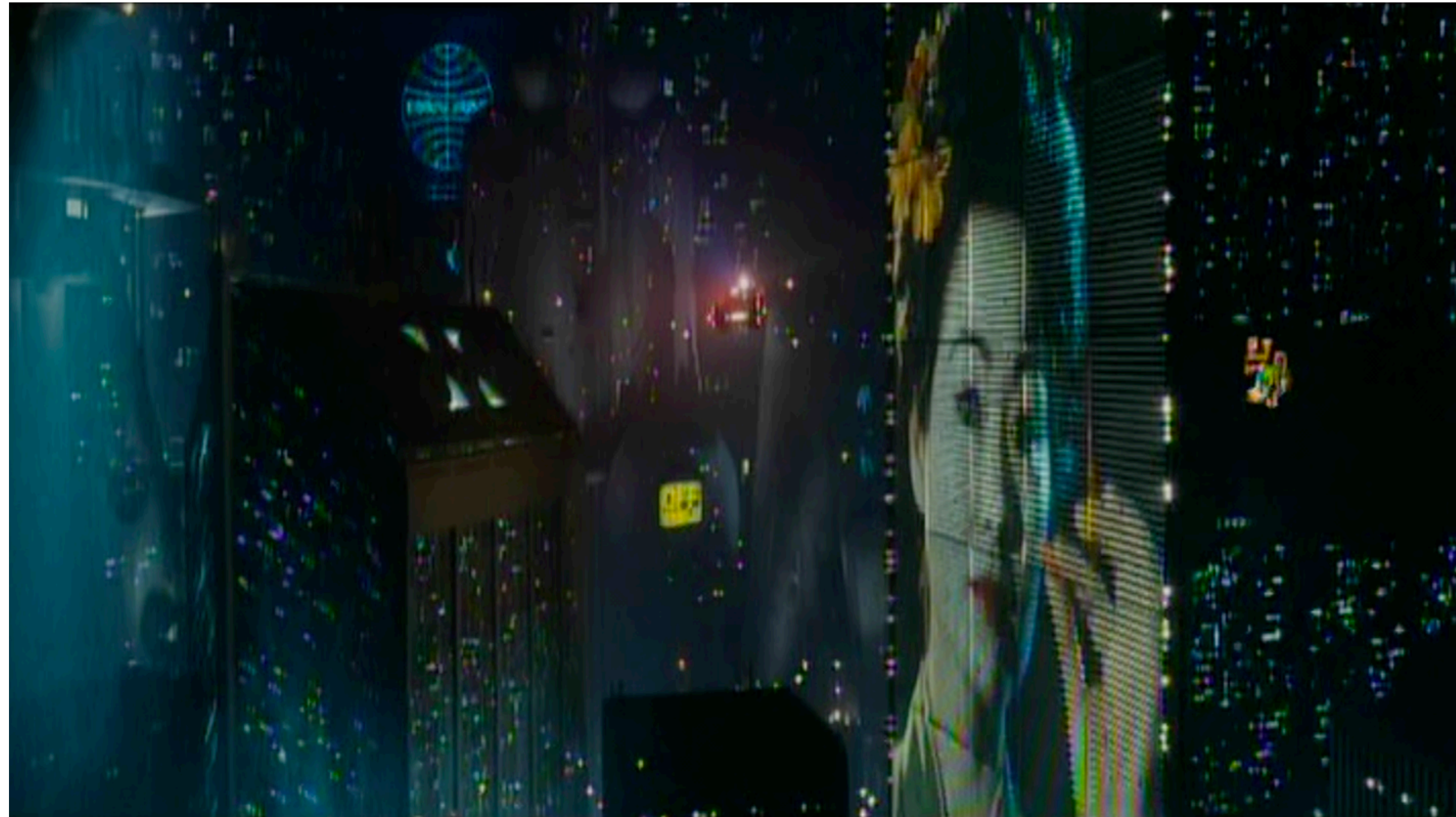
Universal Pictures
(2014)



Is the test that
Caleb is doing with
Ava a Turing Test?

Is emotional
attachment a critical
part of a Turing Test?

Ava is undeniably an intelligent robot, is she also conscious?



Blade Runner

Directed:
Ridley Scott

Warner Brothers
(1982)

The Voight-Kampff test:



How does the Voight-Kampff test differ from the Turing Test?



Blade Runner

Directed:
Ridley Scott

Warner Brothers
(1982)



How are Rachael, Leon,
and Ava different from
each other?

Is seeding Replicants with
memories for capitalistic
gain ethical?

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- ▶ Next Time: Asimov's Laws of Robotics